

# Big Data

# Agenda

- Big Data - Introduction
- Problems
- Solution

# How much you think can happen in 60 secs?

- 400 million \$ sales in Amazon
- 439,000 page views on wiki
- 194,000 apps downloaded from play store
- 31,700 hours of music played
- 40,500 photos uploaded in Insta
- 4.1 million searches on google
- 139,000 hours of video watched in YouTube
- 10 million ads displayed

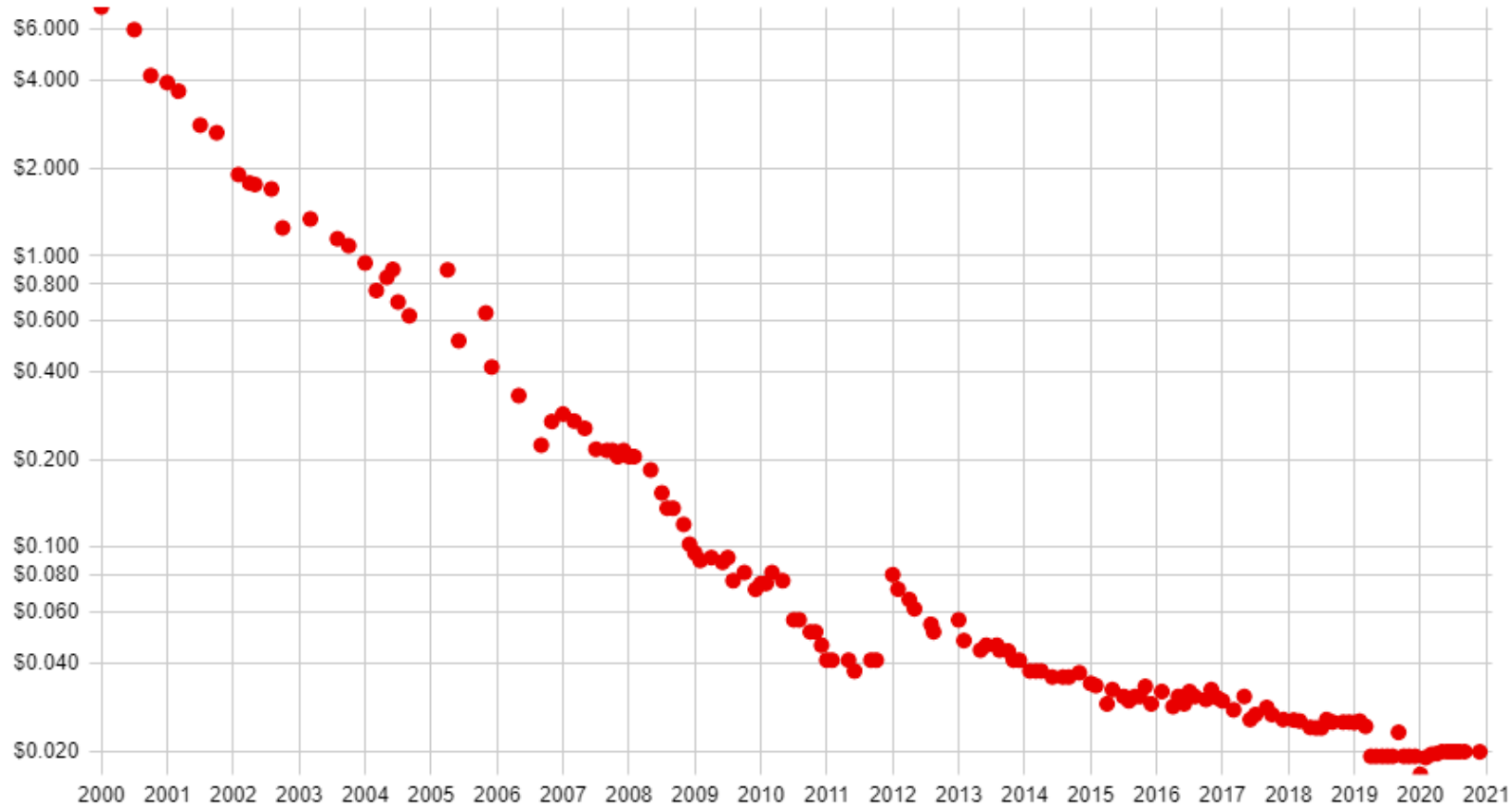
# Petabytes of data every minute (Data Collected)

- 1 Petabyte → 1000 Terabytes
- 1 TB hard disk (1000 such drives are getting filled every minute from web)



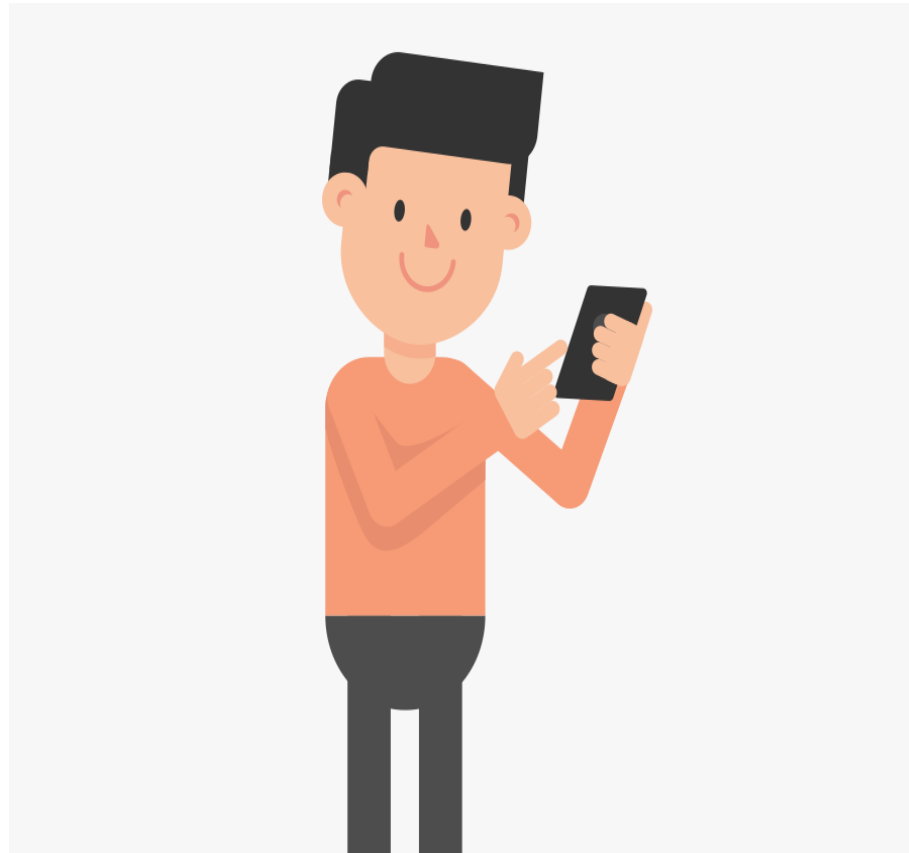
# Because companies can afford !!!!

Dollars per GB of storage over time, log scale



# Data Analogy

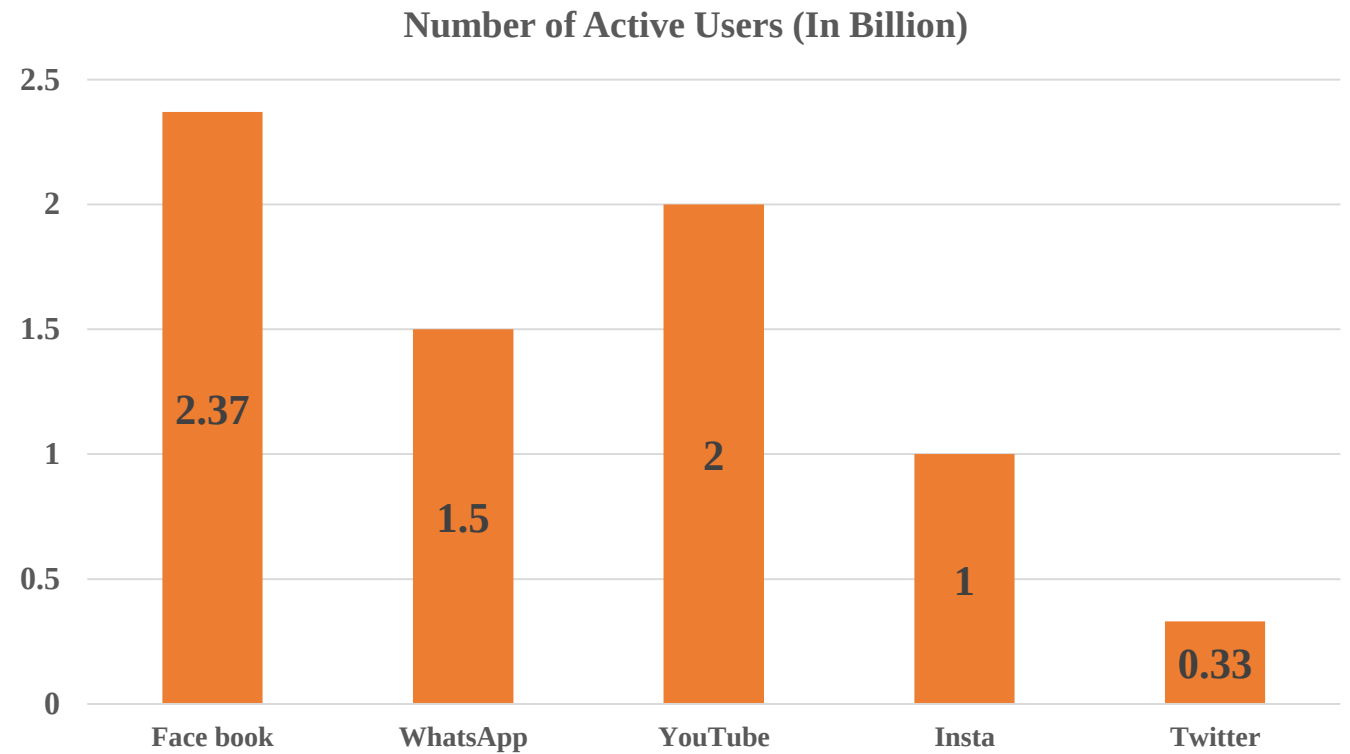
- Gmail
- WhatsApp
- Facebook
- Instagram
- Messenger
- Ola/Uber
- Google pay/ phonepe/ paytm
- Redbus/ Irctc
- Amazon / Flipkart
- YouTube / Hotstar



My name is X  
Age – 25  
Gender – Male

## Few Facts

- Facebook – 2004
- YouTube – 2005
- Twitter - 2006
- WhatsApp – 2009
- Instagram – 2010
- Ola – 2010
- Google Pay – 2018 (55 Million)
- PhonePe – 2015 (218 Million)



# Introduction to Big Data

*What is Big Data?*

*What makes data, “Big” Data?*



# Big Data – In Layman terms



**Lets take one example**

There is a glass with capacity of 250 ML

# Big Data – In Layman terms



**Lets take one example**

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# Big Data – In Layman terms



**Lets take one example**

There is a glass with capacity of 250 ML



248ML -> Data

249ML -> Data

**250ML -> Big Data**

# Big Data – In Layman terms



**Lets take one example**

There is a glass with capacity of 250 ML



248ML -> Data

249ML -> Data

**250ML -> Big Data**

**Lets take another example**

There is a water tank with capacity of 1000L

# Big Data – In Layman terms



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248ML -> Data

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# Big Data – In Layman terms



**Lets take one example**

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998L -> Data

999L -> Data

**1000L -> Big Data**

# Big Data – In Layman terms

Lets take one example

There is a glass with capacity of 250 ML



248ML -> Data

249ML -> Data

**250ML -> Big Data**

Lets take another example

There is a water tank with capacity of 1000L



998L -> Data

999L -> Data

**1000L -> Big Data**

**250ML** is Big Data where as **999L** is still not a Big Data. How?

# Reasons to generate Data (Big Data)





# Reasons to generate Data (Big Data)

- Internet Users



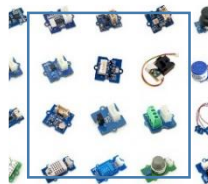
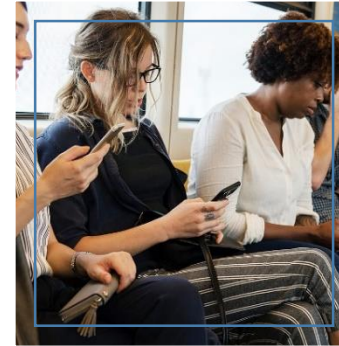
# Reasons to generate Data (Big Data)

- Internet Users
- Smart phones



# Reasons to generate Data (Big Data)

- Internet Users
- Smart phones
- Sensors



# Reasons to generate Data (Big Data)

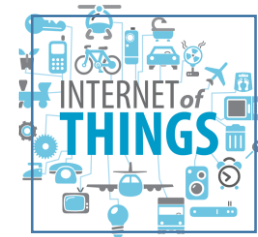
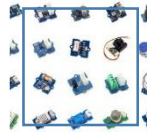
- Internet Users
- Smart phones
- Sensors
- Smart Devices





# Reasons to generate Data (Big Data)

- Internet Users
- Smart phones
- Sensors
- Smart Devices
- IoT

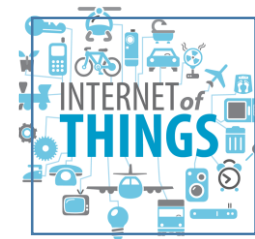


# Reasons to generate Data (Big Data)

- Internet Users
- Smart phones
- Sensors
- Smart Devices
- IoT



- User Friendly Usage



# Big Data Definition

- No single standard definition...

“***Big Data***” is data whose scale, diversity, and complexity require new architecture, techniques, algorithms, and analytics to manage it and extract value and hidden knowledge from it...

# Big Data Definition

- No single standard definition...

“***Big Data***” refers to data that is so large, fast or complex that it’s difficult or impossible to process using traditional methods



# Big Data Definition

- No single standard definition...

“***Big Data***” Big data is a term applied to data sets whose size or type is beyond the ability of traditional relational databases to capture, manage and process the data with low latency. Big data has one or more of the following characteristics: high volume, high velocity or high variety.

- IBM

# Big Data Definition

- No single standard definition...

“***Big Data***” Big data is a just a data which is BIG !!!!

Big Data is something which cannot fit into Traditional database !!!

Big Data is something which takes time than Usual !!!

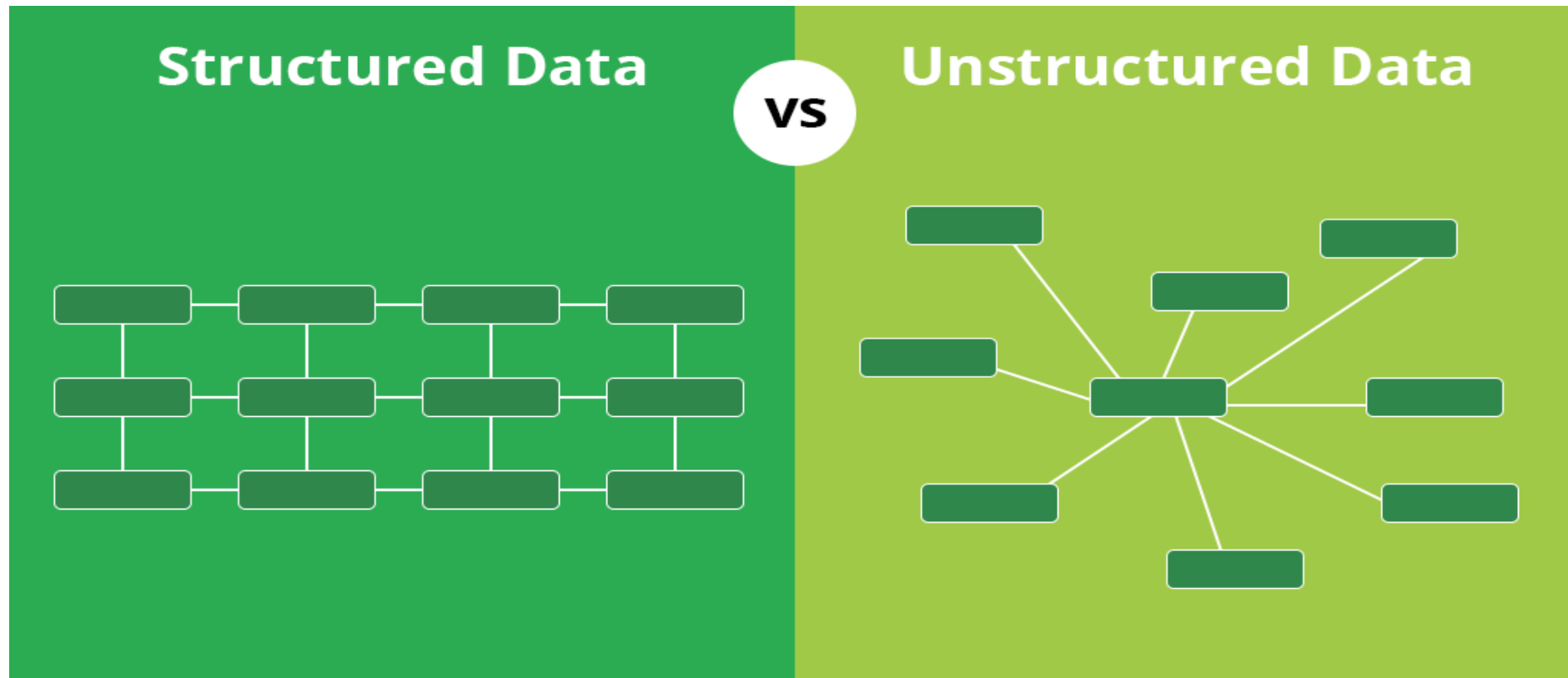
# 5 V's of Big data

## 1. Volume



# 5 V's of Big data

## 2. Variety



# Structured Data

Result Grid





Filter Rows:

Export:



Wrap Cell Content:



|   | actor_id | first_name | last_name    | last_update         |
|---|----------|------------|--------------|---------------------|
| ▶ | 1        | PENELOPE   | GUINNESS     | 2006-02-15 04:34:33 |
|   | 2        | NICK       | WAHLBERG     | 2006-02-15 04:34:33 |
|   | 3        | ED         | CHASE        | 2006-02-15 04:34:33 |
|   | 4        | JENNIFER   | DAVIS        | 2006-02-15 04:34:33 |
|   | 5        | JOHNNY     | LOLLOBRIGIDA | 2006-02-15 04:34:33 |
|   | 6        | BETTE      | NICHOLSON    | 2006-02-15 04:34:33 |
|   | 7        | GRACE      | MOSTEL       | 2006-02-15 04:34:33 |
|   | 8        | MATTHEW    | JOHANSSON    | 2006-02-15 04:34:33 |
|   | 9        | JOE        | SWANK        | 2006-02-15 04:34:33 |
|   | 10       | CHRISTIAN  | GABLE        | 2006-02-15 04:34:33 |
|   | 11       | ZERO       | CAGE         | 2006-02-15 04:34:33 |
|   | 12       | KARL       | BERRY        | 2006-02-15 04:34:33 |
|   | 13       | UMA        | WOOD         | 2006-02-15 04:34:33 |
|   | 14       | VIVIEN     | BERGEN       | 2006-02-15 04:34:33 |
|   | 15       | CUBA       | OLIVIER      | 2006-02-15 04:34:33 |
|   | 16       | FRED       | COSTNER      | 2006-02-15 04:34:33 |
|   | 17       | HELEN      | VOIGHT       | 2006-02-15 04:34:33 |

# Semi-Structured Data

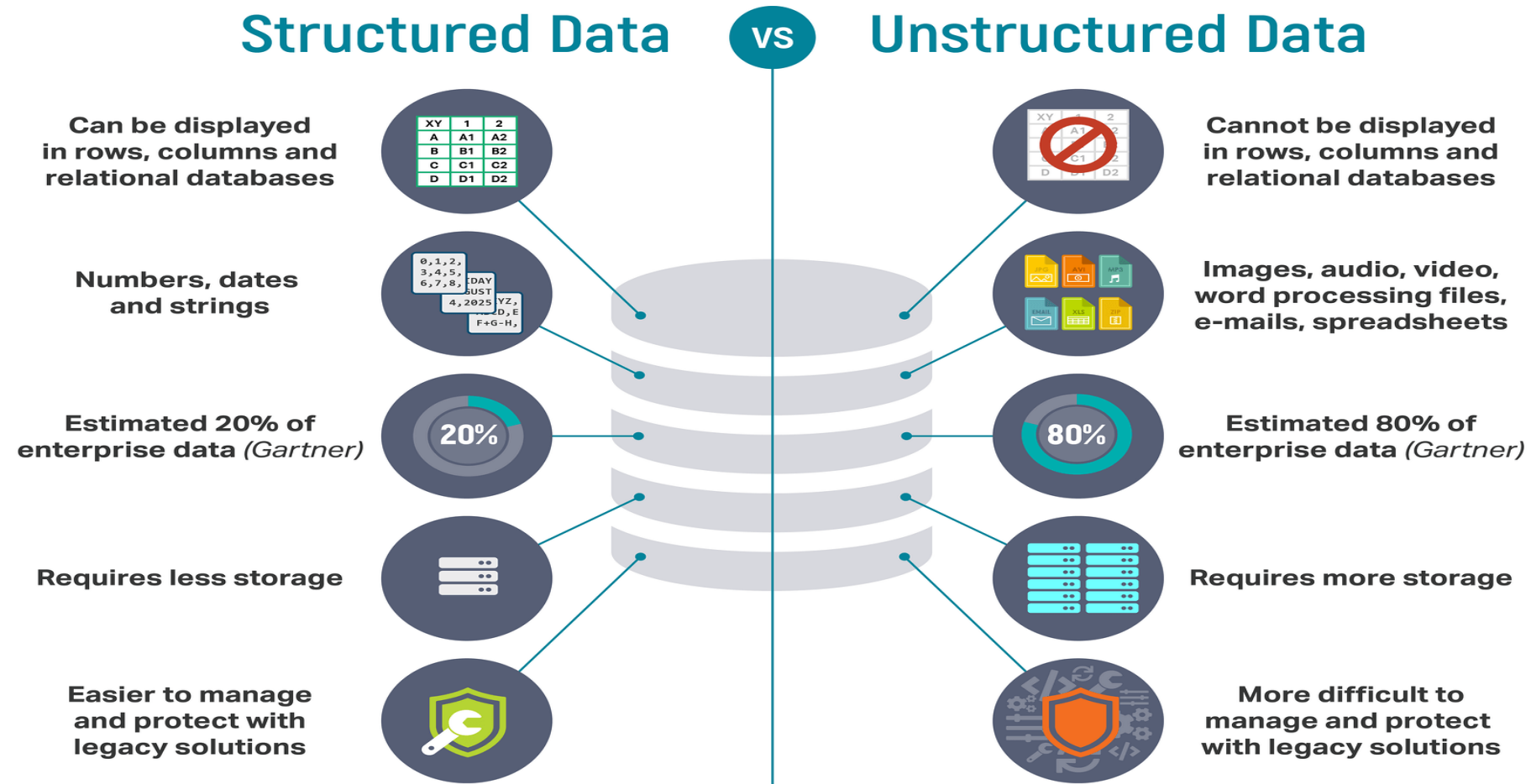
```
09:20:01,054 INFO [org.jboss.modules] (main) JBoss Modules
version 1.3
09:20:01,652 INFO [org.jboss.as.process.Host Controller.status]
(main) JBAS012017: Starting process 'Host Controller'
09:20:05,079 INFO [org.jboss.as.process.Server: myserver.status]
(ProcessController-threads - 10) JBAS012017: Starting process
'Server: myserver'
17:01:58,833 INFO [org.jboss.as.process] (Shutdown thread)
JBAS012016: Shutting down process controller
17:02:03,408 INFO [org.jboss.as.process.Host Controller.status]
(Shutdown thread) JBAS012018: Stopping process 'Host Controller'
17:02:15,246 INFO [org.jboss.as.process.Server: myserver.status]
(ProcessController-threads - 9) JBAS012018: Stopping process
'Server: myserver'
17:03:02,990 INFO [org.jboss.as.process.Server:myserver.status]
(reaper for Server: myserver) JBAS012010: Process 'Server:
myserver' finished with an exit status of 0
17:03:13,170 INFO [org.jboss.as.process.Host Controller.status]
(reaper for Host Controller) JBAS012010: Process 'Host
Controller' finished with an exit status of 0
17:03:13,195 INFO [org.jboss.as.process] (Shutdown thread)
JBAS012015: All processes finished; exiting
```

# Un-Structured Data





# Structured Data vs. Unstructured



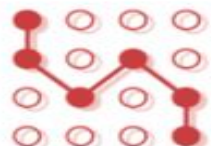


# 5 V's of Big data

## 3. Veracity



Statistical biases



Lack of data lineage



Software bugs



Noise



Abnormalities



Information Security



Untrustworthy  
data sources



Falsification



Uncertainty and  
ambiguity of data



Duplication of data



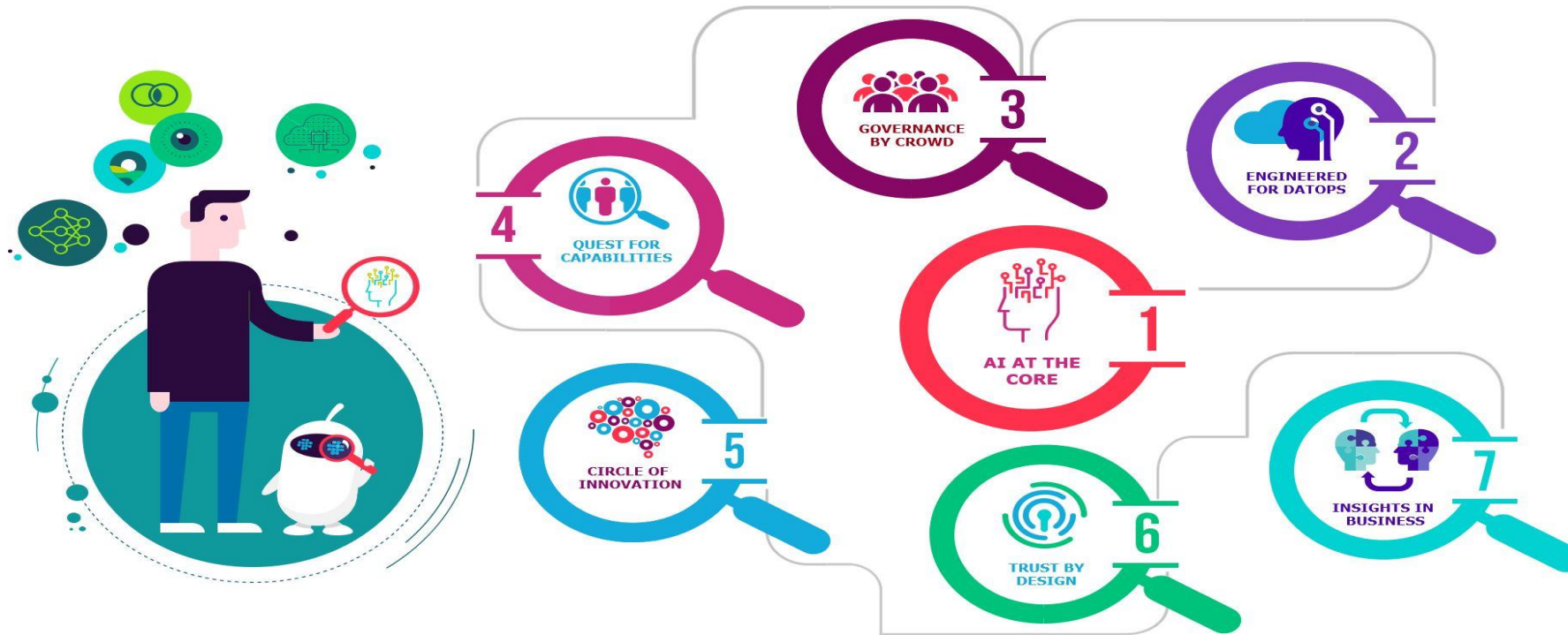
Out of date and  
obsolete data



Human error

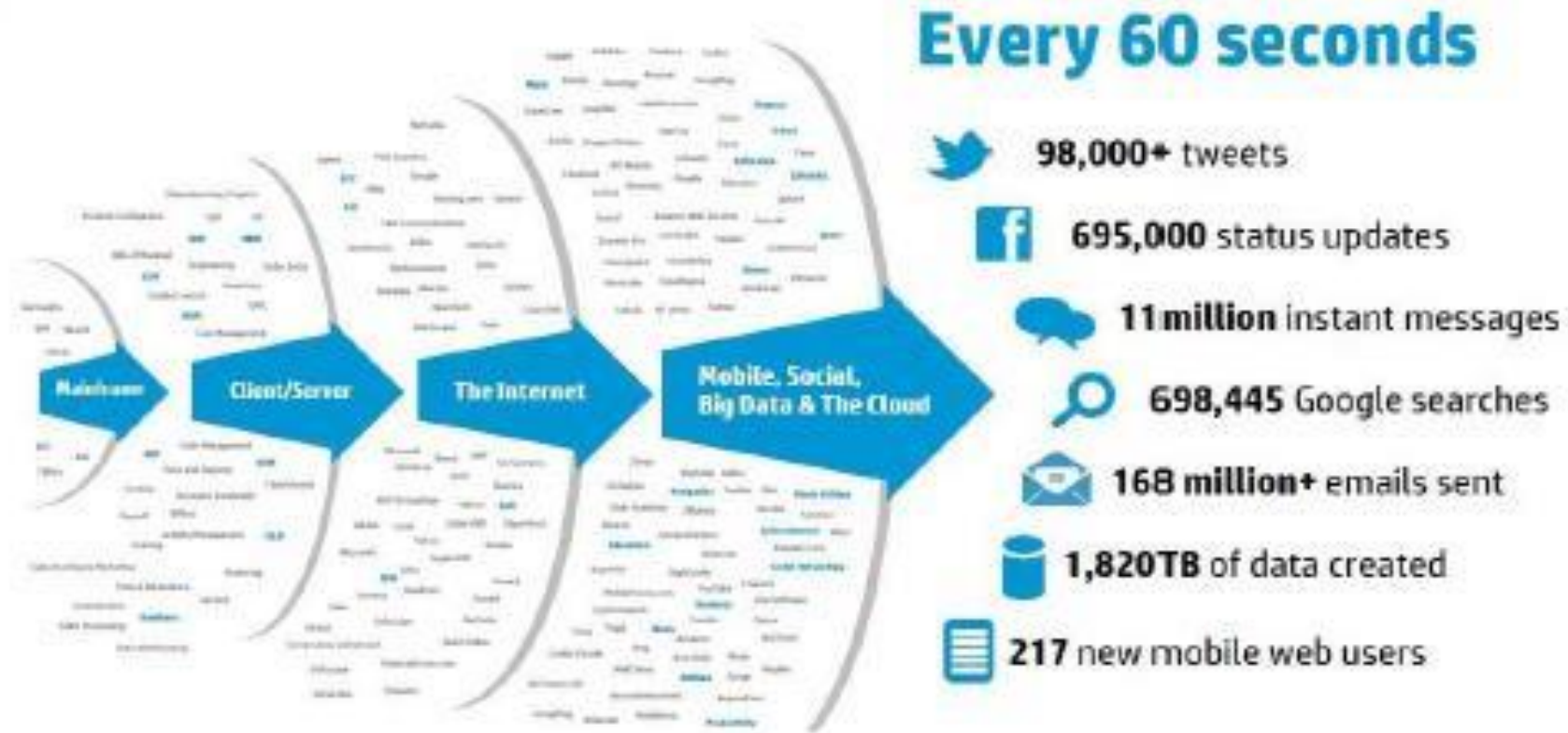
# 5 V's of Big data

## 4. Value



# 5 V's of Big data

## 5. Velocity



# 5 V's of Big data



# Take Away

- Data

# Take Away

- Data
- Big data

# Take Away

- Data
- Big data
- Problems in Big data?

# Take Away

- Data
- Big data
- Problems in Big data?
- Now, What is the solution for big data?



# Take Away

- Data
- Big data
- Problems in Big data?
- Now, What is the solution for big data?



# Questions?

Thank You!!!